

CHE 338: Biochemical Engineering
Unique ID: 16490, 16485
Spring 2026, The University of Texas at Austin

Instructors:

Prof. Brian Belardi (he/him/his), McKetta Department of Chemical Engineering
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Teaching Assistants:

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Tutors:

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Lecture: Tuesdays, Thursdays 9:30-11:00am CT, **CPE 2.218**

Please be punctual and respectful of others. Attendance is *required*, unless you have a doctor's note or a positive viral test result. The lectures will often present unique material not covered by readings or assignments and will offer the opportunity to ask questions in real time. Lectures will be posted to Canvas ~1 week after their recording.

Textbook: *Quantitative Fundamentals of Molecular and Cellular Bioengineering* by Wittrup, Tidor, Hackel, and Sarkar, 1st edition, MIT Press (**optional**)

Online Biochemistry Textbook: [Fundamentals of Biochemistry](#) by Jakubowski and Flatt, 1st edition, LibreTexts (**free**)

Software: *Matlab* (Mathworks, **free for registered UT students**), *ChimeraX* (UCSF, **free**).

Course Website: A course page will be maintained on Canvas – please check it regularly! Course updates and information will be announced through Canvas. Assignments will be posted on Canvas, and course modules will help organize the material.

Course Description and Objectives: The discipline of chemical engineering has transformed the biological sciences, revolutionizing molecular therapies and biotechnology, enabling large-

scale biochemical processes, and establishing the fields of systems and synthetic biology. In this course, students will learn how to analyze complex molecular networks within cells using a quantitative physical and chemical framework. The objective of the course is to turn macroscopic biological events, such as ligand recognition, gene expression, signal transduction, metabolic flux, and protein trafficking, into a mathematically precise and predictive molecular science. Another objective is to provide a comprehensive and concise overview of cell biology and biochemistry relevant to chemical engineers while learning to apply quantitative engineering principles, including thermodynamics, stoichiometry, and particularly kinetics, to the analysis of biological processes. By the end of the course, students will be equipped to propose and analyze new biotechnologies, new therapeutic modalities, and cellular behaviors.

Course Prerequisites:

- One semester of mass and energy balances – ChE317
- One semester of organic chemistry (recommended) – Ch320/Ch328

Grading:

Attendance – 5%

Homework – 20% (lowest score will be dropped automatically)

Quizzes – 15% (lowest score will be dropped automatically)

Group Project – 10%

Midterms Score – 50% (25% each, lowest midterm score will be dropped automatically)

Anticipated course grade cut-offs (greater than or equal to) are the following, A: 93 %, A-: 90%, B+: 87%, B: 83 %, B-: 80%, C+: 77%, C: 73%, C-: 70%, D+: 67%, D: 63%, D-: 60%. Grades may be curved, e.g. these grade cut-offs may be lowered. They will not be raised. Extra credit assignments and points for the entire class may be given at the discretion of the instructor.

Homework: ~Biweekly homework will be assigned for this course and in general will be due ~1 week after posting. *The homework problem sets will reflect concepts and problems that will be tested at a higher level than on exams, i.e. they evaluate different types of questions than exams.* Working in groups is permitted and encouraged, but each assignment must be completed and submitted individually. *You may turn in late, but completed, homework within 1 day of the due day for a 25% reduction in points.* For your homework assignments, please show the detailed steps and logic you used to solve the problem and any code generated as grading will be based on these and not the final answer. We strive to give you points for your effort, but there needs to be evidence of that on the page to justify the points. Remember, we don't know what you are thinking unless you show us explicitly. Completed problems sets will be uploaded to Gradescope.

Reading Assignments: Reading assignments will supplement and complement lecture notes for the semester. Any readings will be announced on Canvas.

Quizzes: Quizzes play an important role in the course and will be administered ~weekly to test conceptual understanding and ensure students are keeping up with the material. Their purpose is to help you evaluate whether you understand and are able to apply the concepts presented in lecture.

Exams: There will be three midterm exams during the semester. The midterms are scheduled for **February 17th, April 2nd, and April 30th.** ***There will be no make-up exams offered since one exam will be dropped from your overall course grade.*** The exams cover material presented

in lectures and on other assignments. Exams will test the skills you acquired in the course. Exam curves may be applied at the discretion of the course instructor and are not guaranteed.

The University of Texas at Austin provides upon request appropriate academic accommodations for qualified students with disabilities. For more information, contact the Disability and Access Office at 571-471-6259. Please contact Prof. Belardi within the first two weeks of class to setup any accommodations.

Project: A group project on translational biotechnology will be assigned. Each group member will receive the same grade for the assignment and will therefore need to contribute equally to the project. A contribution statement for the project will also be included in the final submitted materials.

Course Communication: All course material questions should be addressed during **lecture or office hours**. ***Due to the size of the course, we cannot guarantee that emails/Canvas messages will be answered immediately.*** In the event that you need to communicate about the course electronically, ***please email Prof. Belardi, Prof. Stachowiak, Vinya, and Irene together.*** If you do not include the TA, then a response is not guaranteed.

Re-grade Requests: You have one week after the assignment/exam is returned to you to request a re-grade. Re-grade requests should consist of brief statements that describe your rationale for why a re-grade is necessary on specified problems. A re-grade request will trigger a re-grade of the entire assignment and can result in a lower grade on the assignment/exam.

Absence: No excuses for missed assignments will be accepted other than written certified medical excuses or positive viral test results. By UT Austin policy, you must notify me of your pending absence at least fourteen days prior to the date of observance of a religious holy day. If you must miss a class or a work assignment in order to observe a religious holy day, you will be given an opportunity to complete the missed work within a reasonable time after the absence.

Knowledge, Abilities, and Skills Students Should Gain From this Course:

- Apply material balances to biological processes, such as transcription, translation, signal transduction, protein trafficking and cellular growth
- Apply mass-action chemical kinetics to ligand-protein binding and enzyme activity
- Apply thermodynamics to protein-ligand binding
- Understand structural basis of molecular recognition in biological systems
- Ability to apply numerical methods and computational approaches to kinetic and thermodynamic models and use these to predict future behavior and define design objectives
- An appreciation for the field of biochemical engineering, including the contributions of diverse scientists and their scientific breakthroughs that have had dramatic impacts on humanity
- Ability to work in groups

ABET Student Outcomes for this course:

This course addresses the following ABET student outcomes:

Student Outcome 1: Ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics

Academic Integrity:

The University has a strict policy on academic integrity. Any form of plagiarism or academic dishonesty will NOT be tolerated in ChE 338. If you have any questions, please consult the websites below:

<https://deanofstudents.utexas.edu/conduct/standardsconduct.php>

In particular, students are obliged to follow the Student Honor Code that states “I pledge, as a member of The University of Texas at Austin community, to do my work honestly, respectfully, and through the intentional pursuit of learning and scholarship. I pledge to be honest about what I create and to acknowledge what I use that belongs to others. I pledge to value the process of learning in addition to the outcome, while celebrating and learning from mistakes. This code encompasses all the academic and scholarly endeavors of the University community.” on all quizzes and exams.

Failure to comply with the University’s policy will result in a “zero” in the assignment and an “F” for CHE 338.

Classroom Accessibility and Inclusivity:

The university is committed to creating an accessible and inclusive learning environment consistent with university policy and federal and state law. Please let me know if you experience any barriers to learning so I can work with you to ensure you have equal opportunity to participate fully in this course. If you are a student with a disability, or think you may have a disability, and need accommodations, please contact Services for Students with Disabilities (SSD). Please refer to SSD’s website for contact and more information: <https://community.utexas.edu/disability/>. If you are already registered with SSD, please deliver your Accommodation Letter to me as early as possible in the semester so we can discuss your approved accommodations and needs in this course.

Important Safety Information:

COVID-19 Update: While we will post information related to the immediate situation on campus, you are encouraged to stay up-to-date on the latest news as related to the student experience. <https://healthyhorns.utexas.edu/coronavirus.html>

If you have concerns about the safety or behavior of fellow students, TAs or Professors, call BCAL (the Behavior Concerns Advice Line): 512-232-5050. Your call can be anonymous. Trust your instincts and share your concerns.

Title IX Reporting:

Title IX is a federal law that protects against sex and gender-based discrimination, sexual harassment, sexual assault, sexual misconduct, dating/domestic violence and stalking at federally funded educational institutions. UT Austin is committed to fostering a learning and working environment free from discrimination in all its forms. When sexual misconduct occurs in our community, the university can:

1. Intervene to prevent harmful behavior from continuing or escalating.
2. Provide support and remedies to students and employees who have experienced harm or have become involved in a Title IX investigation.
3. Investigate and discipline violations of the university’s relevant policies. (<https://titleix.utexas.edu/policies>).

Texas Education Code, Section 51.252 requires all employees of Texas universities, including faculty, to report to the Title IX Office any information regarding incidents of sexual harassment,

sexual assault, dating violence, or stalking that is disclosed to them. Texas law requires that all employees who witness or receive information about incidents of this nature (including, but not limited to, written forms, applications, one-on-one conversations, class assignments, class discussions, or third-party reports) must report it to the Title IX Coordinator. Before talking with me, or with any faculty or staff member about an incident of this nature, please be aware that I will be required to report this information. For more information about reporting options and resources, please visit the [Title IX](#) website, contact the Title IX Office via email at: titleix@austin.utexas.edu, or call 512-471-0419.

Personal Pronouns:

Professional courtesy and sensitivity are especially important with respect to individuals and topics dealing with differences of race, culture, religion, politics, sexual orientation, gender, gender variance, and nationalities. Class rosters are provided to the instructor with the student's legal name. I will gladly honor your request to address you by a name that is different from what appears on the official roster, and by the gender pronouns you use (she/he/they/ze, etc.). Please advise me of any changes early in the semester so that I may make appropriate updates to my records. For instructions on how to add your pronouns to Canvas, visit <https://utexas.instructure.com/courses/633028/pages/profile-pronouns>.

Land Acknowledgment:

I would like to acknowledge that we are meeting on Indigenous land. Moreover, I would like to acknowledge and pay my respects to the Carrizo & Comecrudo, Coahuiltecan, Caddo, Tonkawa, Comanche, Lipan Apache, Alabama-Coushatta, Kickapoo, Tigua Pueblo, and all the American Indian and Indigenous Peoples and communities who have been or have become a part of these lands and territories in Texas, here on Turtle Island.

Sharing of Course Materials is Prohibited:

No materials used in this class, including, but not limited to, lecture notes, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have my explicit, written permission. Unauthorized sharing of materials promotes cheating. It is a violation of the University's Student Honor Code. Sites used for sharing materials are frequently monitored, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in initiation of the student conduct process and include charge(s) for academic misconduct, potentially resulting in sanctions, including a grade impact.

Class Recordings:

Class recordings are reserved only for students in this class for educational purposes and are protected under FERPA. The recordings should not be shared outside the class in any form. Violation of this restriction by a student could lead to Student Misconduct proceedings.

Zoom Etiquette:

In case of a Zoom session, please mute yourself unless you are speaking. This will cut down on background noise and limit any distractions for the rest of the class. In general, keeping your video on, if possible, will help you stay focused during the Zoom sessions. As well, feel free to type your question(s) into the Zoom chat such that we can call on you at an appropriate break in class/office hours.

Statement on Flexibility:

In acknowledgement of the recent pandemic and its impact on the University of Texas at Austin community, this course will reaffirm one of the core values here at UT Austin: responsibility. Our responsibility to ourselves and each other is to put our humanity in the forefront of our academic pursuits. With that being said, this semester I commit to being adaptable in time of need. If you experience any hardships such as illness, accident, family crisis please know that these policies may be amended and therefore you should communicate with me as soon as possible. If for any reason you do not feel comfortable discussing with me, please visit Student Emergency Services.

Course Outline:

Note: Changes to the schedule below may be made at my discretion and if circumstances require. It is your responsibility to note these changes when announced in class and via announcements on Canvas.

- I. Cell atlas
- II. Binding equilibria
- III. Biochemical kinetics
- IV. Enzyme kinetics
- V. Network dynamics and metabolism
- VI. Gene expression
- VII. Signal transduction

*Relevant biochemical and cellular structures will be introduced during each unit.